

Interview Problem

feature map
output size

Input Image = $224 \times 224 \times 3$

kernel size = 11×11

num of channels
(RGB)

$$\frac{m - f + 2p}{s} + 1$$

num of kernel = 33

$$\frac{224 - 11}{1} + 1 \Rightarrow 192 \times 192 \times 3$$

① Weights per kernel $\rightarrow 11 \times 11 \times 3 = 363$ Weights

② total Weight for 33 kernel $\rightarrow 363 \times 33 = 11,979$

③ bias (1 per kernel) $\rightarrow 11,979 + 33 \times 1$
 $\Rightarrow 12,012$

TF-IDF $\rightarrow$ frequency

Bag of Words $\rightarrow$ CountVectorizer

Document D1	<u>The child makes the dog happy</u> the: 2, dog: 1, makes: 1, child: 1, happy: 1				
Document D2	<u>The dog makes the child happy</u> the: 2, child: 1, makes: 1, dog: 1, happy: 1				

	child	dog	happy	makes	the	BoW Vector representations
D1	1	1	1	1	2	[1,1,1,1,2] ✓
D2	1	1	1	1	2	[1,1,1,1,2] ✓

Preprocess unique words

key value

frequency

Content

not

taken care of

Word2Vec → Context

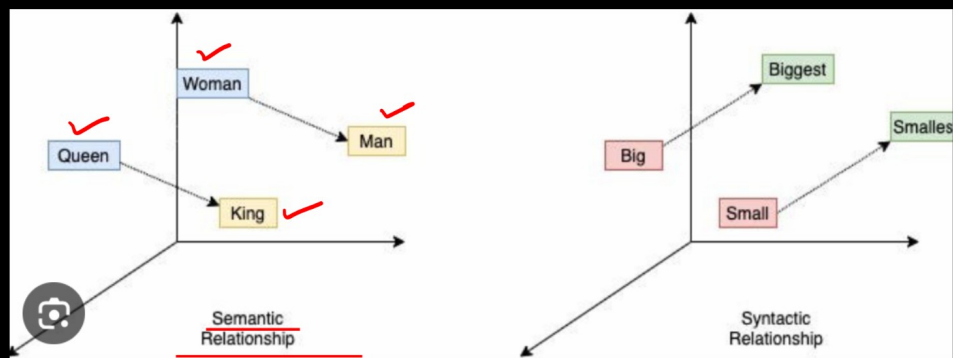
rows

columns (dimensions)

dimensional/features

static embeddings

	is_royal	is_animal	is_male	is_female	is_veg	is_powerful
king	{ 0.95	0.15	0.97	0.04	0.01	0.99}
queen	{ 0.95	0.15	0.04	0.97	0.01	0.99}
man	{ 0.56	0.14	0.98	0.03	0.02	0.50}
woman						
ambiguous						
password						



Word2Vec

similar properties words are very near or comparable to those words which are having no connection

- 512 dimensional
- ① I eat apple daily in the fruit morning → tokenization
- ② I am working in apple → company

Text Preprocessing

maketraw

mapping

a	e	i	o	u
(	(	(	(	(
1	2	3	4	5

h e l l o w o r l d

↓ ↓ ↓

2 4 4